Machine Learning in Finance, ADE+FAB

ASSIGNMENT 3

Analyzing Classification Algorithms for Subscription Prediction

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INTRODUCTION

The objective of the study is to try to build a predictive model as accurately as possible to determine whether a customer will subscribe to a given term deposit. The dataset includes data from 3000 clients with demographic, financial and information about previous marketing interactions with those clients. Accurately predicting the clients' behaviour can help optimise the marketing strategies.

The initial exploration of the dataset shows well-structured data. The dataset has both numerical features(age, balance, duration of last contact) and categorical ones (job type, marital status, education level). It is also important to note the difference in the target variable(y). In this case, the majority of clients did not subscribe to the term deposit. This is important when training and evaluating the different models.

For this, we will conduct different types of analysis on our dataset, including logistic regression models, CHAID, CART, C5.0 and random forest. Through all these analyses, our goal is the super analyst approach and to achieve the greatest possible precision. We will compare all these classification algorithms and identify which one is the most effective and practical when predicting the term deposit subscriptions.

METHODOLOGY

Firstly, we ran a logistic regression, which assigned a coefficient to each predictor variable in order to estimate a client's probability of subscribing to a deposit. As an output, we obtain a linear function transformed by using the logistic function.

Secondly, for the CHAID tree, we applied a CHAID algorithm. This makes decision trees based on chi-squared independence tests, which allow multiway splits and work mainly with categorical values.

Next, we performed a CART tree by training a decision tree with the CART algorithm, forcing binary splits at every node and using the Gini index to determine the purest division.

Later, we ran a C5.0 tree, which selects the splits based on entropy and generates a more complex rule-based structure.

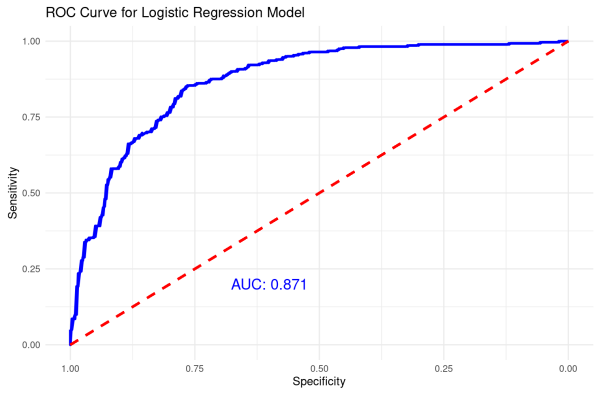
Lastly, we trained a random forest, consisting of a set of decision trees built on different subsamples extracted from the training sample. Each tree uses a random subset of variables at different splits. The final result is obtained by majority voting among the trees.

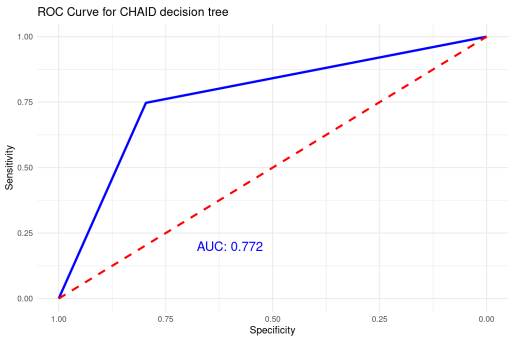
RESULTS

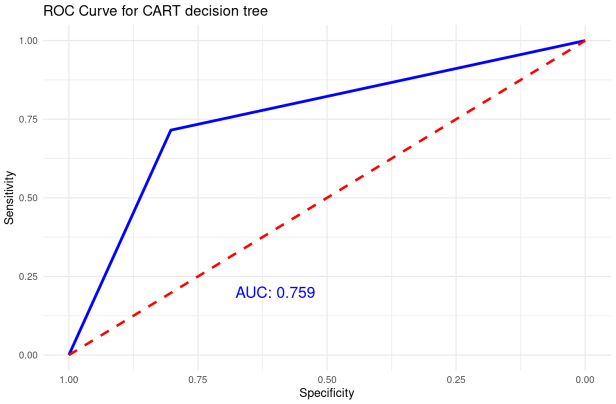
After conducting our analysis as stated above, we have obtained the following results, using in all of them the test sample:

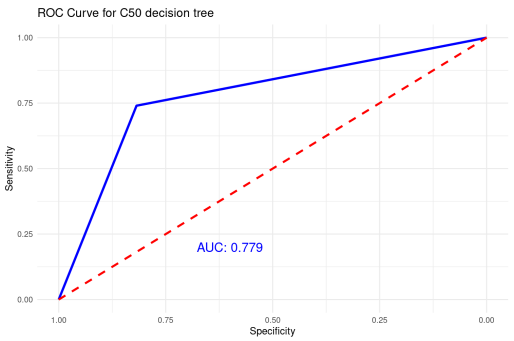
Method	Accuracy	Sensitivity	Specificity	AUC
Logistic regression	0.795	0.8007	0.79	0.8712
CHAID	0.7733	0.7473	0.7962	0.7718
CART	0.7617	0.7153	0.8025	0.7589
C50	0.7817	0.7402	0.8182	0.7792
Random Forest	0.8937	0.8411	0.9365	0.9646

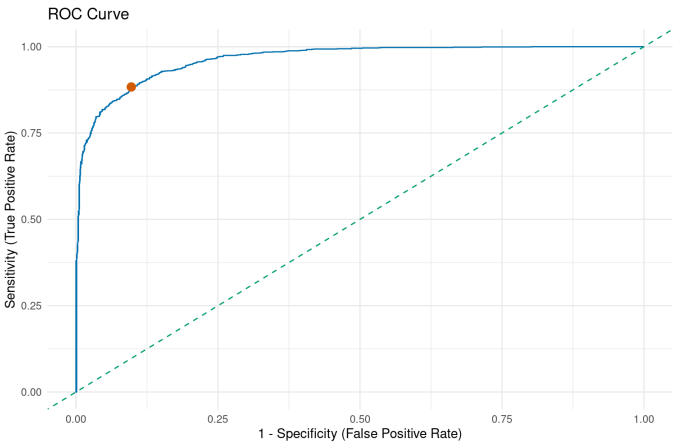
(It should be noted that if we calculate Random Forest's accuracy with the OOB we achieve 0.819)
Additionally, these are the ROC curves for the methods we have tested:



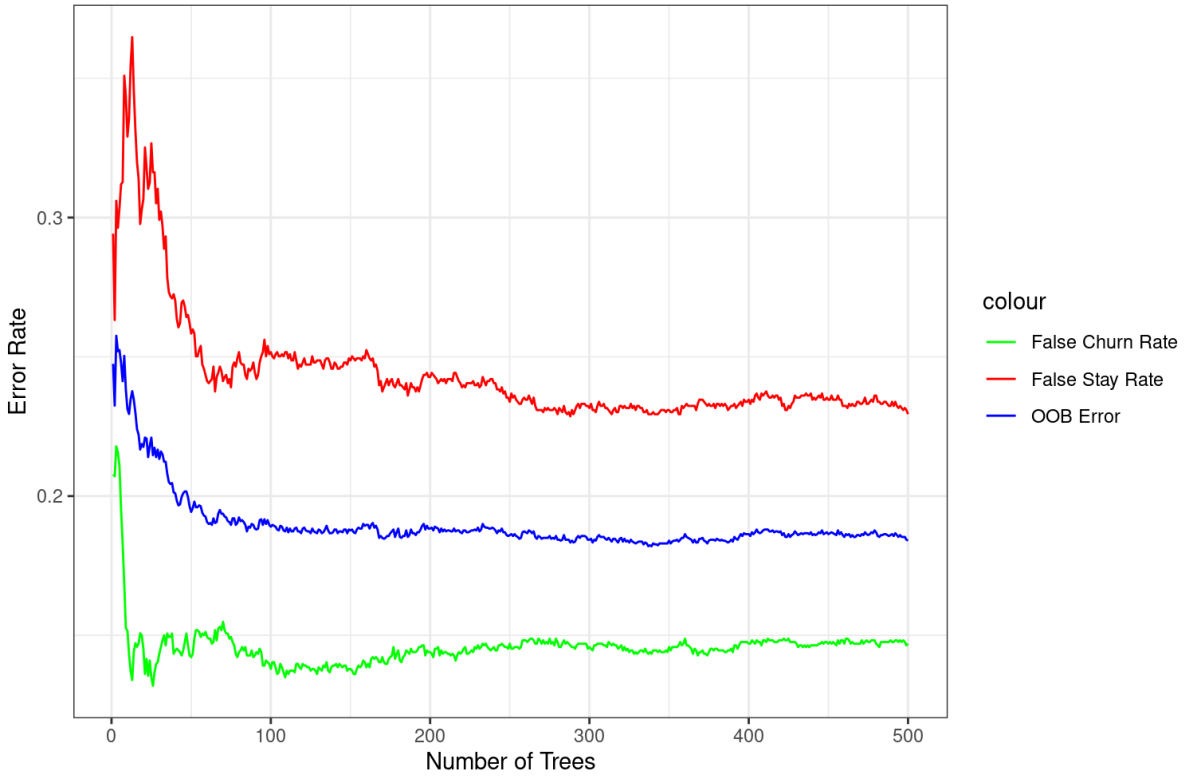








On the other hand, the random forest presents the following error rate depending on the number of trees used:



DISCUSSION & COMPARISON

After thoroughly analysing the results of the different classification models, we can conclude that Random Forest clearly outperforms all other algorithms across every performance metric. It achieves the highest accuracy (89.4%), AUC (0.9646), sensitivity (84.1%), and specificity (93.7%), reflecting strong discriminatory power. Given that our aim is to accurately predict whether a customer will subscribe to the deposit, Random Forest is our recommended algorithm for achieving our goal.

Its strong performance can be attributed to its learning approach: by constructing multiple decision trees on bootstrapped samples and choosing a number of random variables in each tree, Random Forest reduces the risk of overfitting and is less sensitive to noise or outliers. This contrasts with single-tree models, which often lack robustness and generalizability. Consequently, Random Forest stands out as a highly reliable and stable forecasting tool.

It is important to note that the Random Forest model shows lower accuracy when evaluated on the test data compared to its performance on the training set. This can be explained by two main factors. First, the dataset is split into training and test subsets, which reduces the amount of data available for model training. Since Random Forest relies on a large volume of data to construct diverse and robust trees, this reduction can impact its predictive power. Second, Random Forest internally applies bootstrapping during training; it repeatedly samples subsets of the training data to build each tree,

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which introduces additional variability. This process is conceptually similar to the external train-test split, so doing that splitting would be a redundancy.

For this reason, in order to choose the best random forest, we also analyse the out-of-bag (OOB) error, which serves as an internal, unbiased method of accuracy. It provides a reliable estimate of the model's generalisation error without requiring a separate test set, and helps us better understand the expected performance of the model on unseen data. We can use it as an approach of accuracy, and although it is lower than the confusion matrix accuracy (89% vs 82%), it is still the highest accuracy of all models.

Due to its outstanding results, we will consider Random Forest as the benchmark and now focus on comparing the remaining models.

Logistic Regression also demonstrates solid performance, with an accuracy of 79.8% and an AUC of 0.8692. While its predictive performance is not as high as Random Forest's, logistic regression offers a significant advantage in interpretability, allowing us to understand the individual impact of each predictor on the outcome. This makes it a valuable alternative for business contexts where explainability and transparency are key.

Among the three decision tree models—CHAID, C5.0, and CART—the C5.0 algorithm shows the best performance, with an AUC of 0.7792, outperforming both CHAID and CART in most metrics. CHAID follows closely behind, with very similar performance. On the other hand, CART lags behind with the lowest AUC (0.7589) and sensitivity (71.5%), likely due to its constraint of binary splits, which may limit its flexibility and expressiveness compared to CHAID and C5.0. Nonetheless, decision trees may be a great option at the time of showing results clearly for people not familiar with logistic regression.

In summary:

- Best overall performer: Random Forest (high predictive power, suitable for targeting customers).
- Best interpretable model: Logistic Regression (balanced accuracy and explainability).
- Best decision tree: C5.0 (if higher interpretability with tree structures is needed).

Therefore, Random Forest is recommended as the final predictive model, especially in scenarios where maximising predictive accuracy is the main goal. However, it is important to highlight a critical consideration: although Random Forest performs exceptionally well, it operates as a "black box," offering limited interpretability. This lack of transparency may pose challenges when the goal extends beyond prediction to actionable business strategies. For instance, if the marketing team needs to understand which features most influence a customer's decision to subscribe, Random Forest does not provide easily interpretable insights. In such cases, simpler and more interpretable models—such as Logistic Regression or Decision Trees—may be more appropriate. While our primary objective in this study was predictive performance, interpretability remains a valuable criterion, especially in business contexts where understanding the reasoning behind predictions is essential for decision-making.

CONCLUSION

After evaluating several classification algorithms (Logistic Regression, CHAID, CART, C5.0, and Random Forest) we found that Random Forest is the most effective model for predicting whether a client will subscribe to a term deposit. It achieved the best results across all performance metrics, including an accuracy of 89.4%, sensitivity of 84.1%, specificity of 93.7%, and an AUC of 0.9646, showing a strong ability to correctly classify both subscribing and non-subscribing clients. Nonetheless, we should take into account that it is the most expensive method of the previous ones in terms of computing, so this should be something to take into consideration if we are going to be dealing with more data.

Beyond the strong results shown in the metrics, Random Forest is especially useful when the objective is to support the sales process, specifically to decide which clients are worth contacting. Its high sensitivity means that we can detect a large part of the clients who are actually going to subscribe, so we reduce the risk of missing valuable opportunities. At the same time, its high specificity allows us to avoid calling clients who are very unlikely to be interested, making the use of resources more efficient.

Although Logistic Regression and decision trees like C5.0 provide better interpretability and can help explain the influence of different variables, their predictive performance is clearly lower than that of Random Forest. Even though Random Forest proves to be the most reliable and powerful model, in real-world situations, when it comes to marketing analysis or campaign planning (where understanding the reasons behind customer behaviour is essential), other models such as Logistic regression or decision trees offer a clear advantage. Their greater interpretability makes it easier to analyse which factors most influence the client's decision to subscribe. This can be especially helpful when designing more targeted campaigns, identifying market segments, or justifying strategies to stakeholders.

For these reasons, we recommend using Random Forest as the main model for prediction tasks where accuracy and efficiency in client selection are the priority. However, we also believe that simpler and more interpretable models should be used in parallel when the objective shifts toward understanding patterns or building marketing strategies based on insights from the data.

AI use declaration

In this assignment, AI tools have been used in the following ways:
ChatGPT was used to assist with the coding in R, optimizing some functions and helping to debug some errors, and in order to interpret the results, the models and their implications.